

Data Flow Diagram

Field	Details
Team ID	SWTID-2026-7253
Team Size	5
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Project	OptiCrop: Smart Agricultural Production Optimization Engine

Introduction

The Data Flow Diagram illustrates how soil and environmental data flow through the OptiCrop system to generate intelligent crop recommendations.

DFD Symbol Legend

Symbol	Name	Description
External Entity	Farmer/Admin	Interacts with system
Process	ML Processing	Transforms inputs to recommendations
Data Store	Database	Stores user and crop data
Data Flow	Information Flow	Movement between components

Level-0 DFD

Farmer submits soil data (N, P, K, temperature, humidity, pH, rainfall). Administrator manages crop datasets. System returns recommended crop and analytics.

Level-1 DFD

Processes include user authentication, soil data input, preprocessing, machine learning prediction, database storage, crop recommendation, and report generation.

Conclusion

The DFD explains how environmental data moves through OptiCrop to produce accurate crop recommendations and support sustainable agriculture.